

VeriFrame - Technical Documentation

Architecture, Setup Guide & Core Technologies

Technology Stack

Frontend:

- React 19 (Modern UI library)
- TypeScript (Type safety)
- TailwindCSS & shadcn/ui (Premium responsive components)
- TanStack Router (Routing) & Zustand (State management)
- Recharts (Data visualization)

Backend:

- FastAPI (High-performance Python web framework)
- OpenCV & FFmpeg (Video processing, frame extraction)
- NumPy (Statistical and numerical analysis)
- MongoDB Atlas & Motor (Cloud database, async MongoDB driver)
- Cloudinary (Cloud media storage for video & frame uploads)

Deepfake Detection Engine & ML Backend

The detection logic is built on three core pillars:

1. Frame-level Spatial Analysis (examining frames with a CNN)
2. Temporal Consistency Check (analyzing sequence consistency using an RNN)
3. Attention Mechanism (localizing anomalous manipulated regions)

Training Details:

- Datasets: Trained on FaceForensics++, Celeb-DF v2, and DeepFake Detection Dataset.
- Final Accuracy: 85.13% (0.8513)

Backend ML Integration:

The current implementation features simulated scores and mock analysis endpoints. It provides a clean extension point in `python_backend/app/deepfake_model.py` to plug in a real deep learning model's API/ONNX predictions.

Setup and Configuration

Prerequisites:

- Node.js 20+, Python 3.10+, FFmpeg installed and in PATH.

1. Install Frontend Dependencies:

```
cd src/frontend
npm install
```

2. Set Up MongoDB Atlas & Cloudinary Accounts:

Get connection strings and API credentials.

3. Set Up Python virtualenv & Install Requirements:

```
python -m venv venv
source venv/bin/activate (or venv\Scripts\activate)
pip install -r python_backend/requirements.txt
```

4. Configure Environment (.env in python_backend/):

```
MONGODB_CONNECTION_STRING=...
MONGODB_DATABASE_NAME=...
CLOUDINARY_CLOUD_NAME=...
CLOUDINARY_API_KEY=...
CLOUDINARY_API_SECRET=...
```

5. Configure Frontend Environment (.env.local in src/frontend/):

```
VITE_USE_PYTHON_BACKEND=true
VITE_PYTHON_BACKEND_URL=http://127.0.0.1:8000
```

6. Run Servers:

```
Backend: uvicorn app.main:app --reload
Frontend: npm run dev
```

API Architecture & Security

Key Endpoints:

- POST /auth/register: Register a new user
- POST /auth/login: Authenticate and retrieve user token
- POST /analyze: Upload and analyze a video
- GET /analysis/{id}: Retrieve detailed results
- GET /analysis: Get logged-in user's analysis history

Security Controls:

- Password hashing: SHA-256 encryption salted with the username.
- Secure communication: TLS/SSL for database & Cloudinary connections.
- Access control: Endpoint protection forcing user authentication for uploads & logs.